

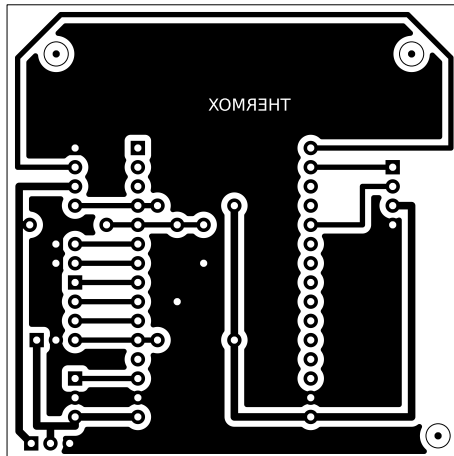
ThermoX PCB: COPPER (60 x 60 mm square, ESP32 rows 22.86 mm apart)

Print at 100 % / Actual size (no 'fit to page'). Check the 50 mm bar with a ruler.

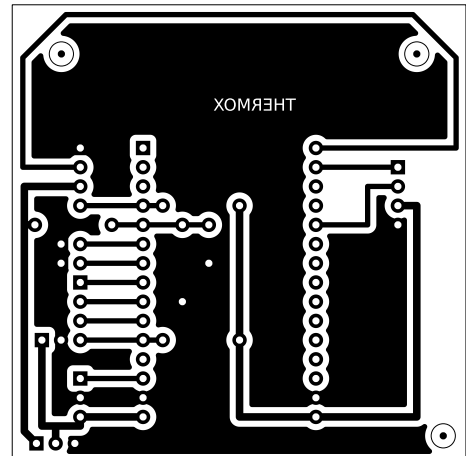
Toner transfer or UV film: use this page as it is. Do NOT mirror it: it is the copper as seen through the board from the parts side, which is the right way round for copper on the bottom.

Check: the small text 'THERMOX' looks mirrored on the paper and reads normally on the etched copper side.

White dots in the pads mark the drill centres: 1.0 mm for headers, 0.8 mm for resistors/capacitor, 3.2 mm mounting holes.



copy 1



copy 2 (spare)

50 mm: measure this after printing. It must be exactly 50 mm.

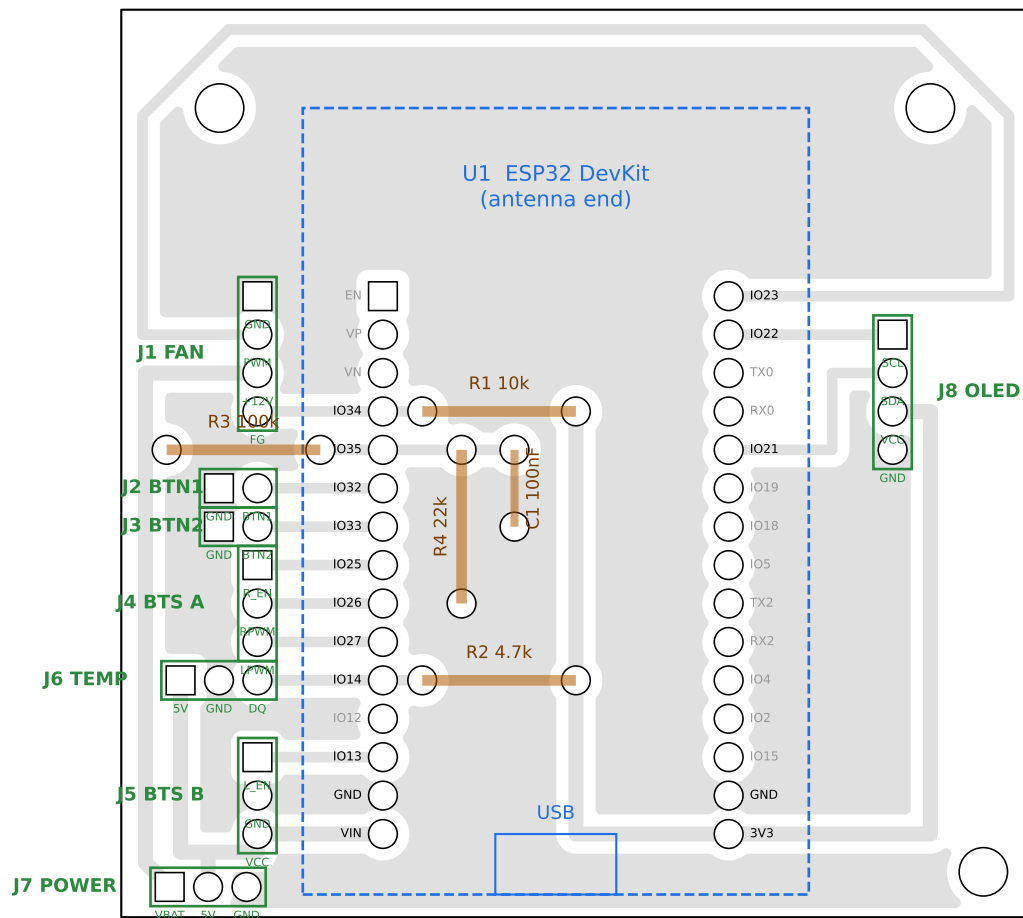


ThermoX PCB: PARTS SIDE (top view, enlarged 2 x)

Parts go on this side; copper is underneath (grey = copper, for reference). Square pad = pin 1.

R1 10k, R2 4.7k, R3 100k, R4 22k and C1 100 nF lie under the ESP32 or the fan header: fit them first, flat on the board.

The ESP32 sits on two 1x15 female headers (USB end at the bottom edge). Fit headers last.



This file: ESP32 pin rows 22.86 mm apart. Measure your ESP32 from the centre of a left pin to the centre of the right pin in the same row: 25.4 mm (1.0") or 22.86 mm (0.9"), and print the matching file.

ThermoX PCB: WIRING, PARTS, NOTES

CONNECTORS

J7 POWER	VBAT: battery + (after the BMS and fuse), also LM2596 IN+. 5V: LM2596 OUT+ (set to 5.0 V first!). GND: battery -, LM2596 IN-/OUT-.
J1 FAN	GND: fan black. PWM: fan blue (control). +12V: fan red. FG: fan yellow/green (tach). Check your SUNON's colours.
J2 / J3 BTN1, BTN2	Each push button: one wire to BTN, one to GND (BTN1 = GPIO32 down, BTN2 = GPIO33 up).
J4 BTS A	R_EN, RPWM, LPWM to the same pins on the BTS7960.
J5 BTS B	L_EN to BTS7960 L_EN, GND to BTS7960 GND, VCC to BTS7960 VCC (5 V logic). R_IS / L_IS stay unconnected.
J6 TEMP	DS18B20: 5V to red (VDD), GND to black, DQ to yellow (data). R2 4.7k pulls DQ up to 3.3 V on the board.
J8 OLED	SCL, SDA, VCC (3.3 V) and GND to the same pins on the 1.5" SH1107 module.

IMPORTANT

Heavy current does NOT go through this board: BTS7960 B+/B- straight from the pack with thick wire (1.5 mm²) and a 10 A fuse, and BTS7960 M+/M- straight to the Peltier.

Battery gauge parts (R3 100k, R4 22k, C1 100 nF on GPIO35) are on the board but the firmware leaves it OFF for now (ENABLE_BATTERY_SENSE = false). After the board is built: BATT_CELLS = 3, BATT_DIVIDER = 5.545, then enable.

Before fitting the ESP32: check with a meter that VBAT, 5V and 3V3 are not shorted to GND, then power J7 with the ESP32 out and check 5 V on J5 VCC and on J6 5V.

PARTS

U1 ESP32 DevKit 30-pin + 2 x 1x15 female header (2.54 mm)

J1 1x4 male header, J2/J3 1x2, J4/J5/J6/J7 1x3, J8 1x4 (or JST-XH of the same pin count)

R1 10 kOhm, R2 4.7 kOhm, R3 100 kOhm, R4 22 kOhm (1/4 W, 1 %), C1 100 nF ceramic

3 x M3 screws/standoffs (square) or 4 (round)

DESIGN CHECK (run by make_pcb.py)

clearance >= 0.5 mm between nets, ground fill 0.8 mm from other nets,
all nets connected, all GND pads on the ground: OK